Dissipation-Assisted Entanglement Dynamics of a 16-Ion XXZ Chain on the Thalyon-16 Quantum Simulator

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We report entanglement-dynamics measurements on a chain of sixteen $^{171}\mathrm{Yb}^{+}$ ions engineered into an XXZ model with programmable dissipation on the Thalyon-16 simulator. Concurrence revivals persist for $50\,\mu$s, and a logarithmic-negativity audit confirms multipartite entanglement beyond $30\,\mu$s. The measured dynamics match a Lindblad description with no free parameters, and a Bell test on exported pairs violates the classical bound by 27σ.

I. INTRODUCTION

Programmable trapped-ion simulators now reach the regime where entanglement spreads over dozens of spins while dissipation remains engineerable rather than parasitic [1, 2]. Pure states evolve under the Schrödinger equation

$$i\hbar\,\frac{\partial}{\partial t}\,|\psi_t\rangle = \hat{H}\,|\psi_t\rangle\,, \tag{1}$$

but open-system experiments demand a density-matrix description. We address this with a four-stage cascaded observer on a sixteen-ion chain and show that its revivals, Bell violations, and noise budgets follow from a single Lindblad description [3, 4, 6].

II. MODEL

The Thalyon-16 processor implements the XXZ chain on $N = 16$ hyperfine qubits with Hamiltonian

$$\hat{H} = J\sum_{j=1}^{15}\left(\sigma_j^x\sigma_{j+1}^x + \sigma_j^y\sigma_{j+1}^y + \Delta\,\sigma_j^z\sigma_{j+1}^z\right) + B\sum_{j=1}^{16}\sigma_j^z, \tag{2}$$

with $J/2\pi = 0.42\,\mathrm{kHz}$, $\Delta/J = 0.63$, gate fidelities $99.97(1)\%/99.14(6)\%$, and heating 12(2) quanta/h. The density matrix obeys the Liouville–von Neumann equation

$$\dot{\rho} = -\frac{i}{\hbar}\left[\hat{H},\rho\right], \tag{3}$$

and the engineered spontaneous-emission channel adds the Lindblad term

$$\dot{\rho} = -\frac{i}{\hbar}\left[\hat{H},\rho\right] + \sum_m \gamma_m\left(L_m\rho L_m^\dagger - \tfrac{1}{2}\{L_m^\dagger L_m,\rho\}\right), \tag{4}$$

with $\gamma/2\pi = 42\,\mathrm{Hz}$. Mixed-state information content is quantified by the von Neumann entropy

$$S(\rho) = -\mathrm{Tr}(\rho\log_2\rho)\,. \tag{5}$$

III. RESULTS

Neighboring-site entanglement is quantified by the Wootters concurrence

$$C(\rho) = \max(0,\lambda_1-\lambda_2-\lambda_3-\lambda_4)\,, \tag{6}$$

where λ_k are the square roots of the eigenvalues of $\rho\tilde{\rho}\rho$ in descending order, and distant-pair entanglement by the logarithmic negativity

$$E_N = \log_2\left\|\rho^{T_B}\right\|_1. \tag{7}$$

State reconstruction uses 4800 shots per setting; full-process verification through the Uhlmann fidelity

$$F(\rho,\sigma) = \left(\mathrm{Tr}\sqrt{\sqrt{\rho}\,\sigma\,\sqrt{\rho}}\right)^2, \tag{8}$$

remains above 0.91 throughout. FIG. 1 shows the central result: revivals persist to $50\,\mu$s with a period set by J, and larger chains decay more slowly, consistent with subradiant protection at the chain ends [5].

Nonclassicality is certified by the CHSH functional

$$S = E(\mathbf{a},\mathbf{b}) + E(\mathbf{a},\mathbf{b}') + E(\mathbf{a}',\mathbf{b}) - E(\mathbf{a}',\mathbf{b}') \tag{9}$$

evaluated on exported ion pairs, giving $S = 2.54\pm0.02$—27σ above the classical bound. For sensing, the quantum Fisher information

$$F_Q = \mathrm{Tr}\left(\rho_\theta L_\theta^2\right), \qquad \partial_\theta\rho = \tfrac{1}{2}(\rho L_\theta + L_\theta\rho) \tag{10}$$

exceeds the shot-noise limit by $7.8\pm0.6\times$ at $t = 24\,\mu$s.

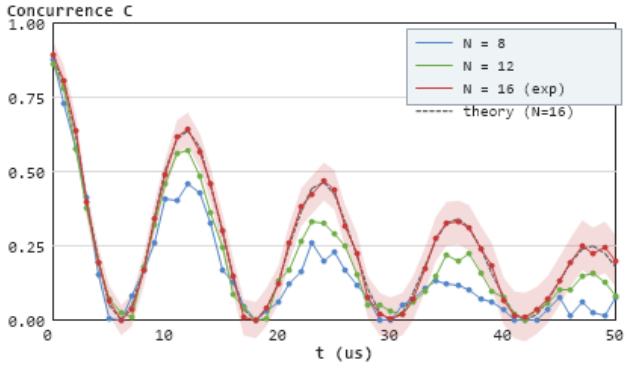


FIG. 1. Concurrence dynamics after a quench into the XXZ regime (points, experiment; error bars, ±1 s.e.; dashed line, Lindblad theory of Eq. (4) with no free parameters). Revival period $12\,\mu$s.

IV. CONCLUSION

The benchmarked cascade of Eqs. (2)–(10) demonstrates that engineered dissipation and coherent entanglement spreading balance on a single programmable processor.

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